

Kazi Samin Mubasshir

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Education

Purdue University

Ph.D. in Computer Science

West Lafayette, IN

Fall 2022 - Spring 2027

- **Advisor:** Dr. Elisa Bertino
- **Research Topic:** Network Security, Cellular Networks, AI for Security
- **Courses:** AI Agentic Security, Security Analytics, Natural Language Processing, Network Security, Information Security, Distributed Systems, Data Communication and Computer Networks, Algorithm Design, Analysis and Implementations

Bangladesh University of Engineering and Technology

B.Sc. in Computer Science and Engineering

Dhaka, Bangladesh

Feb 2016 - Feb 2021

- **Thesis:** Aligner Ensembling, Batch Filtering, and New Datasets for Bengali-English Machine Translation
- **Advisor:** Dr. Rifat Shahriyar
- Graduated with Distinction

Work Experience

Amazon Web Services (AWS)

Applied Scientist Intern

Boston, Massachusetts

Summer [2024, 2025]

- External Security Services (ESS)
- Security Analytics and AI Research (SAAR)

Amazon

Campus Brand Ambassador (CBA)

West Lafayette, IN

Spring 2025 - Present

Purdue University

Graduate Assistantships

West Lafayette, IN

Fall '22 – Spring '26

- **Graduate Research Assistant** (Fall '22 - Fall '24, Spring '26)
 - Cyber Space Security Lab (Cyber2Slab), PI: Dr. Elisa Bertino
- **Graduate Teaching Assistant** (Spring '25, Fall '25)
 - CS180: Problem Solving and Object-Oriented Programming

Advanced Chemical Industries (ACI) Ltd.

Machine Learning Engineer

Dhaka, Bangladesh

Sept 2020 - Aug 2022

- Management Information Systems (MIS)

Research Interests

Security of NextG Networks and Cloud Systems, Cellular Networks, AI for Security, Agentic Security

Research Positions

Purdue University

Graduate Research Assistant

West Lafayette, IN

Fall 2022 - Fall 2024

- Cyber Space Security Lab (Cyber2Slab, PI: Dr. Elisa Bertino)

Bangladesh University of Engineering and Technology

Undergraduate Research Assistant

Dhaka, Bangladesh

Mar 2019 - Feb 2021

- BUET CSE NLP Group (PI: Dr. Rifat Shahriyar)

Publications

CONFERENCE PROCEEDINGS

- Karim, Imtiaz, Hyunwoo Lee, Hassan Asghar, **Kazi Samin Mubasshir**, Seulgi Han, Mashroor Hasan Bhuiyan, and Elisa Bertino. "VWAttacker: A Systematic Security Testing Framework for Voice over WiFi User Equipments." IEEE International Conference on Computer Communications (INFOCOM), 2026.

- Uccello, Federica, Imtiaz Karim, **Kazi Samin Mubasshir**, Elisa Bertino, and Simin Nadjm-Tehrani. "Mission Explainable: From Feature Attribution to Mitigation in 5G Anomaly Detection." International Conference on Computing, Networking and Communication (ICNC), 2026.
- **Kazi Samin Mubasshir***, Imtiaz Karim*, Elisa Bertino, Gotta Detect 'Em All: Fake Base Station and Multi-Step Attack Detection in Cellular Networks, The 34th USENIX Security Symposium (Usenix Sec), 2025, (*joint first authors).
- Imtiaz Karim, **Kazi Samin Mubasshir**, Mirza Masfiquir Rahman, Elisa Bertino, SPEC5G: A Dataset for 5G Cellular Network Protocol Analysis, IJCNLP-AAACL 2023: The 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics
- Abhik Bhattacharjee, Tahmid Hasan, **Kazi Samin**, Md Saiful Islam, M. Sohel Rahman, Anindya Iqbal, Rifat Shahriyar, BanglaBERT: Combating Embedding Barrier for Low-Resource Language Understanding, Findings of the Association for Computational Linguistics: NAACL 2022
- Tahmid Hasan, Abhik Bhattacharjee, Md Saiful Islam, **Kazi Samin**, Yuan-Fang Li, Yong-Bin Kang, M. Sohel Rahman, Rifat Shahriyar, XL-Sum: Large-Scale Multilingual Abstractive Summarization for 44 Languages, Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021
- Tahmid Hasan, Abhik Bhattacharjee, **Kazi Samin**, Masum Hasan, Madhusudan Basak, M. Sohel Rahman and Rifat Shahriyar, Not Low-Resource Anymore: Aligner Ensembling, Batch Filtering, and New Datasets for Bengali-English Machine Translation, Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)

Research Experience

SPEC5G, FBSDetector

Purdue University

Fall 2022 - Spring 2027

- **SPEC5G**: SPEC5G is the first-ever public 5G dataset for NLP research. By leveraging large-scale pre-trained language models that have achieved state-of-the-art results on NLP tasks, we use this dataset for security-related text classification and summarization.
- **FBSDetector**: FBSDetector is an effective and efficient detection solution that can reliably detect FBSes and multi-step attacks from Layer-3 network traces using machine learning at the user equipment (UE) side. To develop FBSDetector, we create the first-ever high-quality and large-scale dataset capturing the network traces in different real-world cellular network scenarios. Our novel machine-learning models, specially designed to detect FBSes and multi-step attacks can effectively detect FBSes with high accuracy and low false positive rates.

Undergraduate Thesis

Bangladesh University of

Engineering and Technology

Mar 2019 - Feb 2021

Not Low Resource Anymore, XLSum, BanglaBERT

- **Not Low-Resource Anymore**: We built a customized sentence segmenter for Bengali and proposed two novel methods for parallel corpus creation on low-resource setups: aligner ensembling and batch filtering. With the segmenter and the two methods combined, we compile a high-quality Bengali-English parallel corpus that was unavailable before.
- **XL-Sum**: XL-Sum is a comprehensive and diverse dataset comprising annotated article-summary pairs. To extract them, we used carefully designed heuristics. As human and intrinsic evaluation indicates, it is highly abstractive, concise, and of high quality.
- **BanglaBERT**: BanglaBERT is a BERT-based Natural Language Understanding (NLU) model pretrained in Bangla. We introduce two downstream task datasets on natural language inference and question answering, and benchmark four diverse NLU tasks covering text classification, sequence labeling, and span prediction. In the process, we bring them under the first-ever Bangla Language Understanding Benchmark (BLUB). BanglaBERT achieves state-of-the-art results outperforming multilingual and monolingual models.

Academic Services

- **Web Chair**
 - NextG Networks Cryptography and Security (NextG-Sec) 2026
 - Workshop co-located with ACNS 2026
- **Journal Reviewer**
 - IEEE Communications Magazine (IEEE COMMAG)
 - IEEE Transactions on Networking (IEEE TON)
- **Journal Sub Reviewer**
 - ACM Transactions on Privacy and Security (ACM TOPS)
- **Conference Sub Reviewer**
 - IEEE Symposium on Security and Privacy (IEEE S&P) 2025, 2026

- ACM Conference on Computer and Communications Security (CCS) 2025
- USENIX Security Symposium (USENIX-Sec) 2026
- Network and Distributed System Security (NDSS) 2024, 2025, 2026
- IEEE European Symposium on Security and Privacy (Euro S&P) 2026
- Applied Cryptography and Network Security (ACNS) 2026
- International Symposium on Research in Attacks, Intrusions and Defenses (RAID) 2023, 2024
- International Conference on Distributed Computing Systems (ICDCS) 2023
- ACM Conference on Data and Application Security and Privacy (CODASPY) 2023
- **Artifact Evaluation Committee (AEC) Member**
 - ACM Conference on Computer and Communications Security (CCS) 2024, 2025
 - USENIX Security Symposium 2023
 - 17th IEEE Workshop on Offensive Technologies (WOOT), co-located with IEEE S&P 2023
- **Graduate Student Mentor**
 - Computer Science Graduate Student Board (CSGSB) Mentor-Mentee Program 2024, 2025
 - Mentored six Purdue CS Graduate Students

Media Coverage and Software Artifacts

- **SPEC5G**: GitHub Repo: <https://github.com/SysNetS/SPEC5G>
- **FBS-Detector**: GitHub Repo: <https://github.com/SysNetS/fbsdetect>
FBSDetector in News: Platforms for Advanced Wireless Research, Purdue Computer Science.

Talks and Presentations

- Gotta Detect 'Em All: Fake Base Station and Multi-Step Attack Detection in Cellular Networks, 34th USENIX Security Symposium, Seattle, Washington, August 2025.
- Detecting Fake Base Stations in Cellular Networks, AWS-SAAR Tech Talks, Amazon, Boston, July, 2024.
- Fake Base Station Detection using PAWR Platforms, Young Gladiators-2023, Northeastern University, Boston, June, 2023.
- Fake Base Station Detection - Summary of research project conducted on the POWDER platform, MERIF-2023, Boston University, Boston, May, 2023.

Awards and Achievements

- 2025 **Travel Award**, Purdue Graduate Student Government (PGSG)
- 2025 **Graduate Travel Award**, Purdue College of Science (CoS)
- 2025 **Student Grant**, USENIX Security Symposium
- 2022 **Robi Datathon 2.0**, Runners-up
- 2022 **AI for Bangla 1.0**, Second runners-up
- 2017-20 **Dean's Listed**, Four consecutive years of the undergraduate studies

Skills

- **Programming Languages**: C/C++, Java, Python, Julia, PHP, Javascript, 8086 Assembly Language, SQL
- **Network Simulators**: Cisco Packet Tracer, Wireshark, NS2, srsRAN, Open5GS
- **Web Frameworks**: Laravel, Django, .NET

Reference

- **Dr. Elisa Bertino** (Ph.D. Advisor)
Samuel D. Conte Professor of Computer Science
Purdue University
- **Dr. Rifat Shahriyar** (Undergraduate Advisor)
Professor
Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology (BUET)

- **Dr. Imtiaz Karim** (Ph.D. Mentor)
Assistant Professor
Department of Computer Science
Erik Jonsson School of Engineering and Computer Science
The University of Texas at Dallas